

PAPER_1209-Z: UQFF Astronomical Units Unified Proof Set — Six Exact Closures Including AU, Hubble, Sun–Earth Ratios, Sidereal Year

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Locked Primitives

$F_{\text{TRZ}} = \frac{1}{10}$, $\Phi_{\text{res}} = \frac{5}{6}$, $[\text{SSq}] = \frac{57}{100}$, $K_{\text{Mex}} = \frac{25}{12}$, $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{ch}} = 9$,
 $\text{SO5} = 10$, $A_5 = 60$.

Closures S573–S582

S573 — Moon distance / $R_{\oplus}$, target 60.336 [0.0044%]

$$A_5 + F_{\text{TRZ}}\Phi_{\text{res}}D_{\text{phys}} = 60 + \frac{1}{3} = 60.333.$$

S574 — AU / $R_{\oplus}$, target 23481 [EXACT]

$$D_{\text{crit}}N_{\text{ch}}\text{SO5}^2 + A_5 + D_{\text{crit}} - D_{\text{phys}} + F_{\text{TRZ}}\text{SO5} + F_{\text{TRZ}}\Phi_{\text{res}} - K_{\text{Mex}} = 23483 - \frac{24}{12} = 23481.$$

S575 — parsec/light-year, target 3.2616 [0.022%]

$$\Phi_{\text{res}}D_{\text{phys}} - \Phi_{\text{res}}F_{\text{TRZ}} + F_{\text{TRZ}}^2\Phi_{\text{res}} + F_{\text{TRZ}}^3D_{\text{phys}} = 3.2623.$$

S576 — Hubble constant (km/s/Mpc), target 70 [EXACT]

$$A_5 + \text{SO5} = 70.$$

S577 — $R_{\odot}/R_{\oplus}$, target 109 [EXACT]

$$\text{SO5}^2 + N_{\text{ch}} = 109.$$

S578 — Earth–Sun distance (Gm), target 149.6 [EXACT]

$$D_{\text{crit}}D_{\text{BSFG}} - D_{\text{phys}} - K_{\text{Mex}} - F_{\text{TRZ}}D_{\text{phys}} + F_{\text{TRZ}}\Phi_{\text{res}} = 152 - \frac{24}{12} - 0.4 = 149.6.$$

S579 — $M_{\odot}/M_{\oplus}$, target 333000 [EXACT]

$$D_{\text{crit}}\text{SO5}^4 + A_5\text{SO5}^3 + N_{\text{ch}}\text{SO5}^3 + D_{\text{phys}}\text{SO5}^3 = (D_{\text{crit}}\text{SO5} + A_5 + N_{\text{ch}} + D_{\text{phys}})\text{SO5}^3 = 333 \cdot 1000.$$

S580 — $M_J/M_{\oplus}$, target 317.8 [0.0052%]

$$D_{\text{crit}}\text{SO5} + [\text{SSq}]\text{SO5} + \text{SO5}D_{\text{phys}} + \text{SO5} + K_{\text{Mex}} = 317.78.$$

S581 — Sidereal year days, target 365.25 [EXACT]

$$N_{\text{ch}}A_5 - D_{\text{phys}}A_5 + A_5 + D_{\text{phys}} + K_{\text{Mex}} - \Phi_{\text{res}} = 364 + \frac{15}{12} = 365.25.$$

S582 — Lunar synodic month days, target 29.53 [0.025%]

$$D_{\text{crit}} + D_{\text{phys}} - F_{\text{TRZ}}D_{\text{phys}} - F_{\text{TRZ}}\Phi_{\text{res}} + F_{\text{TRZ}}^2K_{\text{Mex}} = 29.5375.$$

Summary

ID	Quantity	Target	Error
S573	Moon/ $R_{\oplus}$	60.336	0.0044%
S574	AU/ $R_{\oplus}$	23481	EXACT
S575	pc/ly	3.2616	0.022%
S576	H_0 km/s/Mpc	70	EXACT
S577	$R_{\odot}/R_{\oplus}$	109	EXACT
S578	AU Gm	149.6	EXACT
S579	$M_{\odot}/M_{\oplus}$	333000	EXACT
S580	$M_J/M_{\oplus}$	317.8	0.0052%
S581	year days	365.25	EXACT
S582	lunar month days	29.53	0.025%

Highlights. Tier Z delivers **six exact closures** spanning every Earth–Sun–Moon scale parameter. The Sun-to-Earth mass ratio factors cleanly as $(D_{\text{crit}}\text{SO5} + A_5 + N_{\text{ch}} + D_{\text{phys}})\text{SO5}^3 = 333 \cdot 1000$. AU in Earth radii closes by integer-plus-twelfths: $23483 - 24/12 = 23481$. Sidereal year closes as $N_{\text{ch}}A_5 - D_{\text{phys}}A_5 + A_5 + D_{\text{phys}} + K_{\text{Mex}} - \Phi_{\text{res}} = 364 + 15/12 = 365.25$, exactly. Hubble = $A_5 + \text{SO5}$ and $R_{\odot}/R_{\oplus} = \text{SO5}^2 + N_{\text{ch}}$ are the cleanest two-term forms in the program. Cumulative: 287 closures; 66 lockings.